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NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Sixth Semester B.E. (Civil) (Credit System) Degree Examinations
Supplementary Examinations - December 2025

20CV603 / 17CV603 - TRANSPORTATION ENGINEERING - II

Duration: 3 Hours

Max. Marks: 100

- Note:** 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.
 2) Assume any missing data suitably as per Indian Standards.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Name the different types of gauges adopted in Indian railway lines and indicate the standard dimensions of each as per IRB. Also, explain any four factors to be considered while selecting a suitable type of gauge for railway construction.	10	L*2	1	1
	b) Calculate the maximum permissible train speed for a MG track on a 3° curve. The length of the transition curve is fixed to be 95m. The actual cant is 60mm. The cant deficiency is 51mm. The speed likely to be sanctioned by Railway Authority is 75 kmph.	10	L2	2	2
2.	a) Determine the quantity of the materials required to construct a 2 kilometer stretch of MG railway track using 90 R rails. Assume the sleeper density as N+2.	10	L3	1	1
	b) (i) What do you mean by grade compensation? If the ruling gradient on a BG track section has been fixed as 1 in 200, what should be the compensated gradient when a 4° horizontal curve is to be provided on this gradient? (ii) Explain the various types of rolling resistances that causes resistance to the motion of a train and mention the expression to calculate it.	10	L2	2	1
3.	a) (i) Define creep in rails. Explain Percussion theory of creep with a neat sketch. (ii) List any five requirements of an ideal rail joints.	10	L2	1	1
	b) Calculate the maximum permissible load that can be pulled by a locomotive having four pairs of driving wheels carrying an axle load of 24 tonnes each. The train has to run at a speed of 75 kmph on a straight level B.G track. Also calculate the reduction in speed if the train has to run on a rising gradient of 0.666%. What would be the reduction in speed if the train has to negotiate a curve of radius 572m on the same rising gradient? Assume the coefficient of friction to be 0.166.	10	L3	2	2
Unit – II					
4.	a) Briefly explain the steps involved in following Tunneling methods with the relevant sketches (i) Fore Poling Method (ii) Heading and Benching method.	10	L2	3	1
	b) The length of runway under standard condition is 1620 m. The airport site has an elevation of 270 m. Its reference temperature is 32.90 °C. If the runway is to be constructed with an effective gradient of 0.3%. Determine the corrected runway length.	10	L2	4	2

P.T.O.

5. a) (i) With a neat sketch explain mechanical ventilation of tunnels.
 (ii) On a straight BG track a turnout takes off at an angle of $6^{\circ}42'35''$. Design the turnout when the angle of switch is $1^{\circ}34'27''$. The length of the switch rail is 4.73m. Heel divergence is 11.43 cm and length of the straight arm is 0.85m. 10 L2 3 1
- b) With a neat sketch, explain the various component parts of a single runway airport. 10 L2 4 1
6. a) What do you mean by a Railway station? List and explain any eight factors governing the site selection for an ideal railway station. 10 L2 3 1
- b) (i) Determine the effective gradient for the data given below:
 Take the length of the chain as 20m.
- | | |
|-----------------|--------|
| 0 to 8 chains | +1.0% |
| 8 to 16 chains | -0.8 % |
| 16 to 30 chains | -1.0% |
| 30-40 chains | +0.8% |
| 40-48 chains | +1.1% |
- (ii) What is Regional Planning? Explain the four data to be collected for a scientific and sound regional plan. 10 L2 4 1

Unit – III

7. a) With sketches explain the following:
 (i) Layout of artificial harbor (ii) Mound Break water 10 L2 5 1
- b) Design an exit taxiway joining a runway and a parallel main taxiway. The total angle of turn is 35° and the maximum turn-off speed is 90 kmph. Suggest the maximum separation clearance and draw a neat sketch showing all the design elements. Assume all the missing data suitably as per ICAO/FAA guidelines, and indicate clearly the same at the adequate place. 10 L3 5 2
8. a) Explain any five geometric design elements for taxiway and indicate the typical values as per ICAO specification. 10 L2 5 1
- b) With sketches explain the following:
 (i) Airport Lighting (ii) Wind Direction Indicator. 10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
